Communication-Efficient Semantic Split Federated Learning via Variational Information Bottleneck Driven Dynamic Slicing

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***Abstract*—Split federated learning (SFL) offers a compute-efficient and privacy-preserving solution for resource-constrained Internet of Things (IoT) systems by offloading heavy computations to servers. However, SFL requires transmitting high-dimensional intermediate features at every batch, leading to severe communication bottlenecks and vulnerability to wireless channel noise. Nevertheless, existing approaches have not considered wireless channel state and semantics of those intermediate features jointly during SFL training. Accordingly, this letter proposes a novel channel-aware semantic SFL (CA-SSFL) framework. The client model employs a dual-masking architecture, where a distributed variational information bottleneck (VIB) generates a semantic mask by prioritizing feature dimensions, while a signal-to-noise ratio (SNR)-driven module produces a dynamic channel mask reflecting the transmission budget. The element-wise product of the semantic mask and channel mask enables a dynamic slicing mechanism that suppresses irrelevant or unreliable features under current channel conditions. Extensive experiments on the CIFAR-10 and CIFAR-100 datasets under non-IID settings demonstrate that the proposed CA-SSFL drastically reduces communication costs while outperforming existing methods in both accuracy and communication efficiency under severe noise conditions.**

***Index Terms*—Split Federated Learning, Semantic Communication, Communication Efficiency**

I. Introduction

With the explosive proliferation of Internet of Things (IoT) devices, federated learning (FL) has emerged as a promising paradigm for edge intelligence, enabling collaborative learning while preserving data privacy. However, existing FL methods require each device to learn the entire global model, imposing significant computational and energy burdens on resource-constrained IoT devices [1], [2]. Split federated learning (SFL) [3] addresses this limitation by splitting the model between clients and a central server. By offloading significant computational overhead to the server, SFL dramatically reduces the local computational overhead while maintaining parallel collaborative learning.

However, the computational benefits of SFL come at the cost of a severe communication bottleneck. Unlike FL, which periodically exchanges model weights, SFL requires clients to transmit high-dimensional intermediate feature maps, commonly referred to as smashed data, to the server at every training batch [4], [5]. This frequent exchange of massive data streams generates massive communication overhead and latency, especially over bandwidth-limited IoT wireless networks.

Recent studies have mainly focused on improving communication efficiency in SFL through methods such as quantization, sparsification, and pruning [6], [7]. While these approaches have achieved some success in reducing communication traffic, they fundamentally operate by mechanically compressing data without considering its semantic relevance to the downstream task. For example, magnitude-based pruning or activation sparsification removes low-magnitude activations [8], [9], but this can remove features that might be very important for distinguishing classes, which can significantly reduce inference accuracy, especially at high compression rates. Consequently, conventional approaches fundamentally lack a semantic understanding of the transmitted data. To overcome this limitation, semantic communication has emerged as a practical paradigm shift. Instead of aiming for exact bit-level reconstruction, it focuses on extracting and transmitting only the core semantic information essential for the downstream task [10], [11]. However, integrating these semantic communication principles into SFL frameworks is still in an early stage with limited literature. A few pioneering studies have begun to explore this intersection to address severe communication bottlenecks in distributed edge systems [12], [13].

In particular, Zheng et al. [12] proposed a mobility-aware SFL framework that divides the semantic coder into multiple components to reduce the computational load on vehicles, utilizing transfer learning to efficiently adapt to highly dynamic environments. Building upon this direction, the work of Yu and Chang. [13] introduced a U-shaped SFL architecture equipped with a task-oriented semantic communication module, demonstrating improved robustness against wireless channel noise through an external network status monitor. Despite these advances, existing frameworks still have structural limitations that restrict their practical use in dynamic IoT systems. First, rather than enabling the neural network to adapt to dynamic channel conditions, these frameworks heavily depend on external network-state monitors and complex optimization algorithms to determine compression rates. Second, jointly training the SFL backbone model and semantic module under dynamic channel conditions causes instability; thus, existing methods use two-stage training with a fixed semantic module, which however hinders real-time end-to-end optimization [14], [15]. Third, their effectiveness has not been sufficiently validated under non-independent and identically distributed (non-IID) data settings, which remains a fundamental challenge in realistic SFL scenarios.

To address these limitations, it is essential to enable neural networks to recognize channel states and adaptively compress smashed data. Toward this end, we propose a novel channel-aware semantic split federated learning (CA-SSFL) framework for wireless networks that integrates active channel-aware semantic communication into SFL. To the best of our knowledge, this is the first work to investigate such integration, enabling wireless-adaptive communication overhead reduction. The main contributions are summarized as follows. 1) We propose a new dual-masking architecture that integrates a variational information bottleneck (VIB)-driven Semantic Masking Module and a signal-to-noise ratio (SNR)-driven Channel Masking Module. This enables the client model to evaluate the semantic importance of features and the instantaneous wireless channel state. 2) By integrating these two masks, we introduce a dynamic slicing mechanism. This mechanism selectively reduces irrelevant or unreliable feature dimensions under current channel conditions, thereby enabling autonomous variable-length transmission without relying on an external control module. 3) Extensive experiments on the

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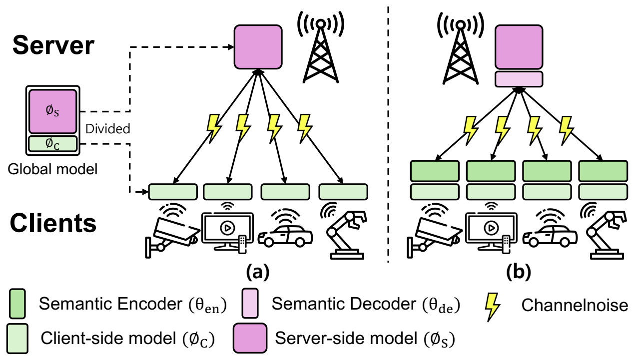


Fig. 1: System model of the proposed scheme.

CIFAR-10 and CIFAR-100 datasets under non-IID settings demonstrate that the proposed CA-SSFL significantly reduces communication overhead while achieving higher accuracy and lower latency than benchmarks.

II. The Overall CA-SSFL Framework

A. System Model

As illustrated in Fig. 1, we consider an SFL framework [3], [4]. In the considered setting, clients communicate with the server over wireless uplink channels. The global model is divided into a client-side sub-model ϕ_c and a server-side sub-model ϕ_s. For an input sample x, the client first computes an intermediate feature representation, commonly referred to as smashed data, as

$$\mathbf{f} = \phi_c(x). \tag{1}$$

The smashed data f is then transmitted to the server through the wireless uplink. The received representation at the server is modeled as

$$\mathbf{f}' = h\mathbf{f} + \mathbf{n}, \tag{2}$$

where $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, \sigma_n^2 \mathbf{I}_K)$ denotes the additive white Gaussian noise (AWGN) vector with noise variance σ_n^2 over the K-dimensional transmitted feature, and h denotes the channel coefficient, with $h = 1$ for the AWGN channel and $h \sim \mathcal{CN}(0, 1)$ for the Rayleigh fading channel.

Based on the received representation, the server performs the remaining layers of the split model to produce the final prediction

$$\hat{y} = \phi_s(\mathbf{f}'). \tag{3}$$

This split architecture reduces the computational burden on client devices by offloading most of the deep neural network computation to the server, which makes it attractive for resource-limited IoT environments. In the generic SFL setting, the split model is trained end-to-end using a standard supervised objective defined on the server output. For a ground-truth label y, the task loss is written as

$$\mathcal{L}_{\text{task}} = \ell(\hat{y}, y), \tag{4}$$

where $\ell(\cdot, \cdot)$ denotes a classification loss such as cross-entropy. During training, the server computes the loss from the final prediction and propagates the intermediate gradient back through the split point so that both ϕ_s and ϕ_c are updated. After local updates, the client-side model parameters are aggregated to form the global model for the next communication round. Although SFL is effective in reducing on-device computation, its communication cost remains substantial because high-dimensional smashed data must be transmitted at every training iteration [5]. This problem becomes more severe when the client-server exchange takes place over wireless channels, where limited bandwidth and channel noise can significantly degrade transmission efficiency and downstream task performance. These challenges motivate the proposed channel-aware semantic transmission mechanism described in the next section.

B. Framework Overview

Building on the SFL framework in Section II, we propose a CA-SSFL architecture to reduce communication overhead while preserving task-relevant information under dynamic channel conditions. Motivated by task-oriented semantic communication [10], [16], the proposed method extends the existing client-server split model by adding a semantic encoder on the client side and a semantic decoder on the server side. Therefore, the client-side model is defined as $\Theta_C = \{\phi_c, \theta_{en}\}$, where ϕ_c extracts smashed data and θ_{en} compresses it into a transmission-efficient latent representation. The server-side model is defined as $\Theta_S = \{\theta_{de}, \phi_s\}$, where θ_{de} reconstructs the received latent representation and ϕ_s performs the downstream prediction task. Therefore, the overall model is given by $\Theta = \{\phi_c, \theta_{en}, \theta_{de}, \phi_s\}$.

Unlike conventional SFL, the proposed CA-SSFL incorporates two key mechanisms in the client-side semantic encoder: 1) a VIB-based semantic masking module that estimates the task relevance of latent features, and 2) an SNR-driven channel masking module that determines their transmission feasibility under the current wireless condition. The final transmission support is determined by combining the semantic mask and the channel mask so that only the feature dimensions that are both semantically important and channel-feasible are selected for transmission. The selected latent features are then sent through a dynamic slicing operation, and the server restores the sparse representation to its original dimensionality before processing it through the semantic decoder and the remaining server-side network. The detailed operations of the semantic encoder, channel masking, dynamic slicing, and end-to-end training objective are presented in the following subsections.

III. Proposed Method

A. VIB-Driven Semantic Masking Module

As shown in (i) of Fig. 2, the flattened intermediate features $\mathbf{f}_{\text{flt}} \in \mathbb{R}^K$ extracted from the client-side model are directly processed by the VIB-Driven Semantic Masking Module. Unlike conventional SFL frameworks that encode inputs as fixed deterministic points, given flattened smashed data $\mathbf{f}_{\text{flt}} \in \mathbb{R}^K$, the encoder outputs a mean vector $\boldsymbol{\mu} \in \mathbb{R}^K$ and a log-variance vector $\log \boldsymbol{\sigma}^2 \in \mathbb{R}^K$ as

$$\boldsymbol{\mu} = g_{\boldsymbol{\mu}}(\mathbf{f}_{\text{flt}}, s), \qquad \log \boldsymbol{\sigma}^2 = g_{\sigma}(\mathbf{f}_{\text{flt}}, s), \tag{5}$$

here, $g_{\boldsymbol{\mu}}(\cdot)$ and $g_{\sigma}(\cdot)$ represent the corresponding encoder mappings, and s represents the current channel state. Then, the latent posterior is defined as

$$p_{\theta_{en}}(\mathbf{z} \mid \mathbf{f}_{\text{flt}}, s) = \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2)). \tag{6}$$

It injects random noise during training to enhance semantic robustness against potential channel distortions. Furthermore, it evaluates the informational value of each feature dimension based on its relevance to the downstream task through dimension-wise Kullback-Leibler (KL) divergence computed from latent statistics modulated by the current channel condition. Finally, this module identifies task-irrelevant dimensions to generate a semantic mask, which will later be fused with the channel masking module to guide the dynamic slicing process.

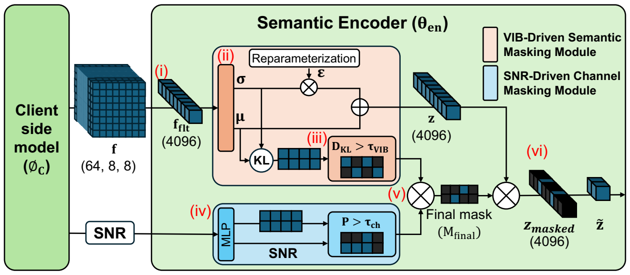


Fig. 2: Proposed semantic encoder.

1) Stochastic Encoding: As shown in (ii) of Fig. 2, the semantic encoder θ_{en} estimates the mean $\boldsymbol{\mu}$ and variance $\boldsymbol{\sigma}^2$ of the latent distribution from $\mathbf{f}_{\text{flt}}$. To enable backpropagation through the stochastic sampling process, we apply the reparameterization trick [17]. The latent vector $\mathbf{z}$ is sampled as

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I), \tag{7}$$

where $\odot$ denotes element-wise multiplication, and ϵ is an auxiliary noise variable sampled from a standard normal distribution. This process internalizes the uncertainty of the wireless channel into the latent space of the deep learning model.

2) Semantic Masking via VIB: As shown in (iii) of Fig. 2, we utilize the dimension-wise KL divergence as a metric for information significance [17]. Let K denote the total latent dimension, and let $k \in \{1, \ldots, K\}$ denote the index of a latent component. Since the posterior is modeled as a diagonal Gaussian, the posterior distribution of the k-th latent component is given by $p_{\theta_{en}}(\mathbf{z}_k \mid \mathbf{f}_{\text{flt}}, s) = \mathcal{N}(\mathbf{z}_k; \mu_k, \sigma_k^2)$, while the prior is defined as $r(\mathbf{z}_k) = \mathcal{N}(\mathbf{z}_k; 0, 1)$. The KL divergence for the k-th latent dimension is then calculated as

$$D_{\text{KL}}^{(k)} = -\frac{1}{2}\left(1 + \log \sigma_k^2 - \mu_k^2 - \sigma_k^2\right). \tag{8}$$

Here, $D_{\text{KL}}^{(k)}$ denotes the KL divergence between the posterior and prior distributions of the k-th latent component. Note that $D_{\text{KL}}^{(k)}$ is computed analytically from the posterior parameters (μ_k, σ_k^2) rather than from a sampled latent realization $\mathbf{z}_k$. Therefore, the auxiliary noise variable ϵ in (7) does not explicitly appear in the KL expression. During the forward pass, dimensions with a $D_{\text{KL}}^{(k)}$ value strictly greater than a predefined semantic threshold τ_{VIB} are considered highly informative. Based on this evaluation, the module generates a binary semantic mask $M_{\text{VIB}} \in \{0, 1\}^K$:

$$M_{\text{VIB}}^{(k)} = \mathbb{I}\left(D_{\text{KL}}^{(k)} > \tau_{\text{VIB}}\right), \tag{9}$$

where $\mathbb{I}(\cdot)$ is the indicator function. This semantic mask M_{VIB} is passed to the subsequent fusion stage to be jointly considered with the instantaneous wireless channel constraints.

B. SNR-Driven Channel Masking Module

In wireless networks, the available communication capacity fluctuates severely due to dynamic channel conditions. To address this physical constraint, we introduce the SNR-Driven Channel Masking Module, which acts as an active physical gatekeeper to dynamically allocate the transmission budget.

As shown in (iv) of Fig. 2, this module converts the instantaneous SNR into a dynamic transmission threshold bounded by $\tau_{\min}$ and $\tau_{\max}$. Let

$$s_{\text{norm}} = \frac{s - s_{\min}}{s_{\max} - s_{\min}}, \tag{10}$$

where s denotes the instantaneous SNR, and $s_{\min}$ and $s_{\max}$ denote the minimum and maximum SNR values used for normalization, respectively. Then, the channel threshold is defined as

$$\tau_{\text{ch}} = \tau_{\max} - s_{\text{norm}}(\tau_{\max} - \tau_{\min}), \tag{11}$$

where $\tau_{\max}$ and $\tau_{\min}$ denote the upper and lower threshold bounds, respectively. Thus, $\tau_{\text{ch}} \in [\tau_{\min}, \tau_{\max}]$, approaching $\tau_{\max}$ under poor channel conditions and $\tau_{\min}$ under favorable channel conditions.

A multilayer perceptron (MLP) processes s_{norm} to generate a channel-specific probability vector $P \in [0, 1]^K$, where K is the total dimension of the latent representation. Each element $P^{(k)}$ represents the estimated transmission feasibility of the corresponding latent dimension under the current channel condition. The MLP is not trained from an explicit supervision target on P; instead, it is updated with the entire CA-SSFL model through the end-to-end training objective. Because P determines the channel mask and thereby affects dynamic slicing, semantic reconstruction, and the final task prediction, the MLP is updated indirectly through the end-to-end training objective, which encourages channel-feasibility scores leading to a favorable accuracy-communication trade-off.

Finally, the binary channel mask $M_{\text{ch}} \in \{0, 1\}^K$ is obtained by applying a gating mechanism to each element $P^{(k)}$ of the probability vector P using the dynamic threshold τ_{ch}:

$$M_{\text{ch}}^{(k)} = \mathbb{I}\left(P^{(k)} > \tau_{\text{ch}}\right), \tag{12}$$

From (12), when the channel quality is poor (i.e., $s_{\text{norm}} \to 0$), τ_{ch} approaches $\tau_{\max}$, which makes the channel mask more selective. In contrast, when the channel quality is favorable (i.e., $s_{\text{norm}} \to 1$), τ_{ch} approaches $\tau_{\min}$, allowing more latent dimensions to remain active. Operating in parallel with the VIB-driven Semantic Masking module, this SNR-driven Channel Masking module translates SNR-based channel constraints into a binary mask, filtering out transmission-unviable features.

C. Dual-Masking, Dynamic Slicing, and Loss Function

1) Dual-Masking and Dynamic Slicing: To jointly account for both semantic importance and channel feasibility, we combine the semantic mask and channel mask into a final transmission mask. As shown in (v) of Fig. 2, the final mask is defined as

$$M_{\text{final}} = M_{\text{VIB}} \odot M_{\text{ch}}, \tag{13}$$

where $\odot$ denotes element-wise multiplication. This formulation preserves a latent dimension only when it is both semantically informative and feasible for transmission under the current channel condition. The sampled latent vector $\mathbf{z}$ is then masked as

$$\mathbf{z}_{\text{masked}} = \mathbf{z} \odot M_{\text{final}}. \tag{14}$$

After this masking step, we apply a dynamic slicing operation that transmits only the active latent entries and their corresponding indices. Let

$$\mathcal{A} = \{k \mid M_{\text{final}}^{(k)} = 1\} \tag{15}$$

denote the active index set.

However, transmitting $\mathbf{z}_{\text{masked}}$ directly does not reduce the communication cost, as the zero-valued elements still occupy the bandwidth. As shown in (vi) of Fig. 2, the transmitted latent vector $\tilde{\mathbf{z}}$ is then obtained by dynamically slicing $\mathbf{z}_{\text{masked}}$, i.e., by retaining only the entries of $\mathbf{z}_{\text{masked}}$ indexed by $\mathcal{A}$ together with their corresponding indices. At the server side, the sparse latent representation is restored to its original dimensionality by allocating a zero-initialized tensor and placing the received values back to their original positions using the transmitted indices before semantic decoding.

TABLE I: Performance analysis of CA-SSFL on CIFAR-10 AWGN (**B**: Best, U: Second)

β	τ_{VIB}	Acc. (%)	Comm. (GB)	τ_{max}	τ_{min}	Acc. (%)	Comm. (GB)
0.100	1.5	20.34	8.22	0.7	0.2	**41.71**	14.03
	1.0	39.41	12.63		0.3	40.70	14.05
	0.5	40.20	15.51		0.4	41.39	13.41
0.050	1.5	16.57	**8.19**	0.8	0.2	40.44	13.23
	1.0	**41.39**	13.41		0.3	39.80	12.85
	0.5	39.82	17.39		0.4	41.53	12.52
0.010	1.5	32.38	8.66	0.9	0.2	40.08	12.64
	1.0	40.46	15.87		0.3	40.83	12.05
	0.5	37.66	24.05		0.4	41.27	**11.55**
0.005	1.5	32.69	9.27				
	1.0	40.83	17.12				
	0.5	40.69	31.12				
0.001	1.5	39.04	9.63				
	1.0	40.98	20.59				
	0.5	38.20	39.41				

*** Fixed Settings**
- Left: $\tau_{max} = 0.7$, $\tau_{min} = 0.4$
- Right: $\beta = 0.05$, $\tau_{VIB} = 1.0$

TABLE II: Comparison of Accuracy and Communication Cost under AWGN and Rayleigh Channels

Method	**AWGN Channel**		**Rayleigh Channel**	
	Acc. (%)	**Comm. (GB)**	**Acc. (%)**	**Comm. (GB)**
SFL	**51.04**	150.07	**47.43**	150.07
SC-USFL	36.69	20.68	38.64	20.68
CA-SSFL (Ours)	41.71	**14.03**	40.41	**14.59**
w/o Semantic Masking	37.66	45.43	38.67	18.12
w/o Channel Masking	39.34	17.63	38.19	17.90

TABLE III: Comparison of Computational Load and Model Size

Algorithm	**MFLOPs**			**Model Size (MB)**		
	Client	**Server**	**Total**	**Client**	**Server**	**Total**
SFL	19.76	50.34	70.10	0.57	42.05	42.62
SC-USFL	23.26	53.84	77.10	1.95	43.43	45.38
CA-SSFL (Ours)	35.16	78.20	113.36	2.03	42.89	44.92

To address the instability and gradient vanishing issues caused by the extreme sparsity of dynamic slicing, we employ sparsity-aware gradient compensation. Specifically, the back-propagated gradients from the server are dynamically scaled by the inverse of the active feature ratio ($1/\max(p, 0.01)$). This scaling mechanism, inspired by the principle of inverted dropout [18], ensures that the expected magnitude of the learning signals remains consistent regardless of the fluctuating transmission budget.

2) Loss Function Design: The proposed CA-SSFL is designed to ensure high accuracy in downstream tasks under a given channel state s while reducing the amount of transmitted smashed data for communication efficiency. To achieve this, we adopt the Information Bottleneck (IB) principle [19]. We formulate the loss function as the total loss $\mathcal{L}_{\text{Total}}$, which consists of the task-specific loss $\mathcal{L}_{\text{Task}}$ and the information compression loss $\mathcal{L}_{\text{Compression}}$, balanced by a hyperparameter $\beta > 0$, which governs the trade-off between task performance and communication data compression rate. Depending on β, the model reduces the information of features irrelevant to the task:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{Task}} + \beta \mathcal{L}_{\text{Compression}}. \tag{16}$$

Here, $\mathcal{L}_{\text{Task}}$ measures the downstream prediction error, while $\mathcal{L}_{\text{Compression}}$ regularizes the latent representation toward a compact prior. Using the variational upper bound of the VIB framework [20], these two terms are instantiated as

$$\mathcal{L}_{\text{Task}} = \mathbb{E}_{\mathbf{z} \sim p_{\theta_{en}}(\mathbf{z}|\mathbf{f}_{\text{flt}}, s)} \left[-\log q_{\theta_{de}, \phi_s}(y \mid \tilde{\mathbf{z}}') \right], \tag{17}$$

and

$$\mathcal{L}_{\text{Compression}} = D_{\text{KL}}\left(p_{\theta_{en}}(\mathbf{z} \mid \mathbf{f}_{\text{flt}}, s) \,\|\, r(\mathbf{z}) \right), \tag{18}$$

where $p_{\theta_{en}}(\mathbf{z} \mid \mathbf{f}_{\text{flt}}, s)$ denotes the encoder posterior distribution, $q_{\theta_{de}, \phi_s}(y \mid \tilde{\mathbf{z}}')$ denotes the predictive distribution of the decoder and server-side model, $D_{\text{KL}}(\cdot\|\cdot)$ denotes the KL divergence, and $r(\mathbf{z}) = \mathcal{N}(\mathbf{z}; 0, I_K)$ is the corresponding standard multivariate Gaussian prior over the latent vector. By reducing this divergence, the encoder is penalized for transmitting excess information. This enables the encoder to capture most important semantic features using fewer dimensions.

IV. Performance Evaluation

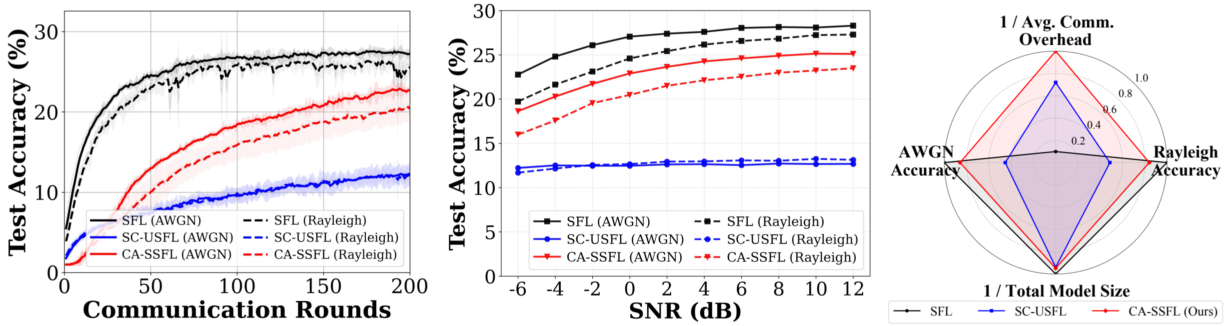
A. Experimental Setup

We evaluate the proposed method using the CIFAR-10 and CIFAR-100 dataset in AWGN and Rayleigh fading channels. To simulate a non-IID environment, each client was assigned a local dataset of 3,000 non-overlapping training images. To construct the non-IID setting, we partition the dataset across clients using a Dirichlet distribution with concentration parameter $\alpha = 0.5$. We employ a split ResNet-18 architecture, where the client processes the first layer and the server handles the remaining layers 2, 3, and 4. For semantic communication, the widely validated DeepJSCC architecture [21] serves as the backbone. The following baselines were established for comparison: 1) SFL [3] transmits raw smashed data without semantic compression. 2) SC-USFL [13] is a state-of-the-art semantic SFL approach utilizing a pre-trained semantic module and an external Network Status Monitor (NSM). We train all models over a continuous SNR range of [−5, 15] dB to evaluate their adaptability to dynamic wireless environments. All results were calculated by averaging 4 random seeds.

B. Performance Comparisons

1) Hyperparameter Sensitivity Analysis: Table I shows the trade-off between classification accuracy and communication overhead of the proposed CA-SSFL, analyzing the effects of the VIB parameters (β, τ_{VIB}) and the dynamic threshold bounds (τ_{max}, τ_{min}). Regarding the VIB module (left), increasing the semantic threshold τ_{VIB} strictly reduces the communication overhead. Interestingly, lower τ_{VIB} values (e.g., 0.5), which permit more data transmission, do not guarantee better accuracy. In fact, setting the threshold near $\tau_{VIB} = 1.0$ or 1.5 can reduce communication overhead while maintaining classification accuracy. This indicates that moderately aggressive slicing effectively filters out task-irrelevant noise and prevents overfitting in non-IID settings, whereas transmitting redundant features degrades global convergence. Furthermore, the right-hand side of the table illustrates the effect of channel-aware dynamic thresholds. The optimal balance was achieved by fixing $\beta = 0.05$ and $\tau_{VIB} = 1.0$, and setting $\tau_{max} = 0.7$ and $\tau_{min} = 0.2$. We performed the same experiment on the Rayleigh channel, fixing $\beta = 0.1$ and $\tau_{VIB} = 1.0$, and applying $\tau_{max} = 0.7$ and $\tau_{min} = 0.4$. This confirms that adaptively constraining the transmission budget based on instantaneous SNR significantly balances the rate-distortion trade-off.

2) Performance Comparison with Benchmarks: Table II examines communication efficiency and robustness. On the AWGN channel, SFL achieves the highest accuracy of 51.04% but at the cost of a very large communication overhead of 150.07 GB. SC-USFL reduces the communication cost to 20.68 GB, but with a much lower accuracy of 36.69%. By comparison, the proposed CA-SSFL achieves 41.71% accuracy with only 14.03 GB of communication overhead,



(a) Accuracy vs. Round (b) Accuracy vs. SNR (c) Efficiency–Accuracy Trade-off

Fig. 3: Performance comparison on CIFAR-100 dataset under Non-IID conditions: (a) test accuracy vs. communication rounds, and (b) test accuracy vs. SNR, and (c) Efficiency–Accuracy Trade-off Radar

demonstrating a better accuracy-communication trade-off than both baselines. As illustrated in Fig. 3(a), the proposed CA-SSFL maintains convergence, overcoming performance degradation under non-IID and low-SNR conditions. As shown in Fig. 3(b), test accuracy is evaluated across various SNR levels. SFL becomes increasingly vulnerable at low SNR, as high-dimensional raw feature transmission is more easily corrupted by severe channel noise. Since SC-USFL selects one of four fixed compression ratios based on the channel conditions, it maintains stable performance but suffers from significantly reduced accuracy. In contrast, CA-SSFL demonstrates graceful degradation and maintains superior accuracy over SC-USFL even under low-SNR conditions. This proves that the proposed channel-aware dynamic slicing effectively protects essential semantic information by filtering out noise-sensitive dimensions. Furthermore, as shown in Fig. 3(c), the proposed CA-SSFL achieves a better trade-off between model size, communication overhead, and accuracy.

3) Ablation Study: Finally, the ablation results in Table II confirm the importance of the proposed dual-masking architecture under both channel types. Removing the VIB-based Semantic Masking module causes a larger degradation than removing the SNR-based Channel Masking module in both AWGN and Rayleigh channels. In AWGN, the accuracy drops from 41.71% to 37.66% and the communication overhead increases from 14.03 GB to 45.43 GB without semantic masking, whereas removing channel masking leads to a smaller accuracy drop to 39.34% and a more moderate increase in communication overhead to 17.63 GB. A similar trend is observed in Rayleigh, where removing semantic masking reduces the accuracy to 38.67% and increases the communication overhead to 18.12 GB, while removing channel masking results in 38.19% accuracy and 17.90 GB communication overhead. These results indicate that semantic masking plays the primary role in filtering out task-irrelevant latent features, whereas channel masking provides an additional channel-aware selection mechanism that further improves the accuracy-communication trade-off under varying channel conditions.

4) Computational Overhead Comparison: Table III compares the computational load (MFLOPs) and model size (MB) of various models to evaluate the feasibility of CA-SSFL in resource-constrained IoT environments. Compared to standard SFL, CA-SSFL introduces an additional client-side overhead of 15.40 MFLOPs and 1.46 MB to operate the dual-masking semantic encoder. Such an overhead is practically acceptable in modern IoT devices and represents a worthwhile trade-off, since wireless bandwidth and spectrum are far more constrained resources than local computation. Hence, CA-SSFL exchanges a modest increase in computation for a substantial reduction in communication overhead. This trade-off is also more favorable than that of SC-USFL, which incurs additional model complexity but still yields lower accuracy and higher communication cost than the proposed method.

V. Conclusion

In this letter, we have proposed a novel CA-SSFL framework that effectively balances the trade-off between communication efficiency and task performance via channel-aware dynamic slicing. The proposed CA-SSFL can be particularly beneficial in resource-constrained IoT scenarios, where low communication overhead, bandwidth efficiency, and robustness against dynamic wireless channels are critical.

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